Learning Incomes

- Familiarity with the techniques of mathematical induction and strong mathematical induction.

6. **Proof Reading.** Consider the following claim and the following proof.

   Claim: The largest natural number is 1.

   Proof. Let N be the largest natural number: Therefore for every $n \in \mathbb{N}$ we have $N \geq n$. Since $1 \in \mathbb{N}$, we know $N \geq 1$. And since $N^2 \in \mathbb{N}$, we know $N \geq N^2$. Therefore $N^2 - N \leq 0$. So $N(N-1) \leq 0$. Thus $N - 1 \leq 0$. And so $N = 1$.

   Therefore, if N is the largest natural number, then $N = 1$. □

   The claim is is false, but all of the logical steps in the proof seem to be reasonable. Explain.

7. **Mathematical Induction: Equalities and Inequalities**

   Using a proof by induction, prove the following statements.

   (a) $(\forall n \in \mathbb{N}) \quad 0^3 + 1^3 + 2^3 + 3^3 + \cdots + n^3 = \frac{1}{4}n^2(n+1)^2$

   (b) $n^2 \leq n!$ whenever $n \geq 4$

8. **Powers of i.** Using a proof by induction and the fact that $i^2 = -1$, prove that for all $n \geq 0$ we have

$$\begin{aligned} i^{4n+1} &= i \\ i^{4n+2} &= -1 \\ i^{4n+3} &= -i \\ i^{4n+4} &= 1 \end{aligned}$$

   (Yes, one could prove this directly using the power laws for complex numbers. But proving things that we already know to be true with a new method helps us better understand the new method.)

9. **The Fibonacci Numbers**

   The Fibonacci Numbers is the sequence of natural numbers that starts with 0 and 1 and proceeds so that each subsequent number is the sum of the previous two:

$$0, 1, 1, 2, 3, 5, 8, 13, 21 \ldots$$

   This sequence of numbers is named after an 12th Century Italian mathematician. Fibonacci studied these numbers as a pretense to solving a rather unrealistic applied problem of counting the number of rabbits after a fixed number of generations of breeding. Beyond the Fibonacci numbers, Fibonacci is best known for creating a mathematics textbook named *Liber Abaci*.

   Though this sequence of numbers is named for Fibonacci, he is far from the first to have studied them. References to this sequence appear as far back as India in 200 BCE. At this time, these numbers were being examined by an Indian author named Pingala. He used these numbers to count the number of fixed-length music patterns that could be made using short and long syllables. Pingala's work on short and long syllable sequences can be seen as an early form of binary notation.

   Despite the fact that Pingala was studying these numbers more than 1000 years before Fibonacci was born, we have canonized Fibonacci in the literature and left the contributions of Pingala in the margins.

   For $n \geq 0$, let f_n denote the nth Fibonacci number. That is, let $f_0 = 0, f_1 = 1, f_2 = 1, f_3 = 2$, etc... By definition we have $f_n = f_{n-1} + f_{n-2}$ for all $n \geq 2$

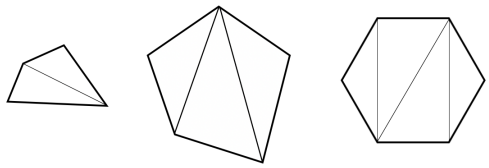
   Using a proof by strong induction prove:

$$f_n = \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n$$

   for all $n \geq 0$.

10. Challenge. **Dividing Polygons.** Let $n \geq 4$ be an integer.

Consider the process of fully dividing up a convex polygon with n sides using non-intersecting lines from vertex to vertex. For example here are fully divided convex polygons with $n = 4, 5, 6$ sides.



Can't remember what a convex polygon is? Google it...

(a) Find a formula for the number of regions in a fully divided convex polygon with n sides.

(b) Prove that your formula is correct using a proof by induction. It might help to label the vertices of your n-gon with the labels $1, 2, 3, \ldots, n$.

(c) Consider replacing *convex polygon* with *regular polygon* in part (a). Why doesn't your proof from (b) still work?